Make7Robotics

Grade 1 → Grade 9

The Make7Robotics Learning Journey

From Thinking to Creating

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What Should a Student Be Able to Do by Grade 9?

The 10 Cognitive steps

By Grade 9, students master ten critical cognitive and technical milestones, transitioning seamlessly from consuming technology to engineering independent solutions.

- **01.** THINK
- **02.** IDENTIFY
- **03.** DECOMPOSE
- **04.** DESIGN
- **05.** BUILD
- **06.** TEST
- **07.** DEBUG
- **08.** IMPROVE
- **09.** PRESENT
- **10.** CREATE

Progression Arc



- Explorer
- Thinker
- Builder
- Programmer
- Developer
- **Independent Creator**

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Our Educational Philosophy

"Coding is the vehicle.

Thinking is the destination."

Our curriculum is intentionally designed around 5 core pillars to empower learners from fundamental cognitive logic to real-world technology production.

Cognitive Thinking
Building mental models for complex logic.

Problem Solving
Breaking down challenges into actionable steps.

Tinkering
Hands-on exploration and mechanical understanding.

Technology Creation
Moving from consumption to production.

Entrepreneurship
Applying solutions to real-world market needs.

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The 9-Year Development Path

A progressive capability framework for cognitive growth and technological mastery

GRADE 1
EXPLORE
Foundational logic and spatial awareness

GRADE 2
DISCOVER
Visual sequencing and patterns

GRADE 3
THINK
Algorithmic reasoning

GRADE 4
BUILD
Modular construction and early mechanics

GRADE 5
PROGRAM
Syntax and formal abstraction

GRADE 6
CREATE
Multi-domain integration

GRADE 7
SOLVE
Advanced system architecture

GRADE 8
DEVELOP
Application engineering

GRADE 9
INNOVATE
Independent product creation

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Age-Appropriate Learning & Cognitive Depth

"Same ecosystem. Different cognitive depth."

Our framework scales seamlessly across grade levels, introducing foundational computational thinking early and progressing into sophisticated software architecture and end-to-end engineering.

GRADE 2

Visual Logic & Sequencing

Building the foundational mental model through structured visual paradigms and algorithm flows.

GRADE 5

Python & Syntax Logic

Transitioning from visual blocks to text-based coding and core data structures.

GRADE 8

App Dev & Full Stack

Architecting integrated application systems, user interfaces, and robust backend logic.

GRADE 9

Advanced Systems

Independent, end-to-end engineering, complex algorithms, and autonomous system design.

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Grades 1–2 Curriculum Stage

EXPLORE

Foundational Computational Mindsets

Cognitive Objective

Building spatial, computational, and sequential mental models before introducing formal syntax.

Core Focus Areas

Pattern Recognition & Sequencing
Identifying structures, logic loops, and ordered steps.

Visual Logic & Creativity
Translating abstract ideas into visual problem-solving modules.

Visual Coding
Tangible interaction with block-based inputs and outputs.

Tinkering & Interactive Storytelling
Hands-on exploration through narrative-driven tech challenges.

Simple Game Mechanics

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GRADES 3–4 STAGE

THINK & BUILD

Cognitive Objective

Mastering problem decomposition and establishing logical rule structures for complex tasks.

Students transition from basic operational literacy to systematic computational logic, constructing integrated physical and digital solutions.

Computational Logic & Software

- Algorithms, loops, and conditional statements
- Game development fundamentals

Systems Thinking & Engineering

- Problem decomposition and debugging
- Modular robotics and mechanics

Digital Projects & Web

- Introductory web concepts and digital projects
- Applied rule-based project execution

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Grade 5 Stage

PROGRAM

Cognitive Objective:

Bridging visual thinking into formal syntax and abstraction, enabling precise programmatic control.

Text-Based Programming

Transitioning to formal language structures. Core Python fundamentals including variables, conditions, loops, and functions.

Algorithm Design & Debugging

Developing foundational algorithm design principles combined with systematic debugging to analyze and solve errors logically.

Application Building

Applying syntax mastery to construct simple, functional applications synthesizing algorithmic thought into execution.

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Grade 6 Curriculum Stage — CREATE

Cognitive Objective: Executing multi-domain building and designing interactive software systems.

CORE FOCUS 01

Intermediate Python Development

Advanced logic structures, data manipulation, modular programming, and algorithm optimization.

CORE FOCUS 02

Web Development Foundations

Building interactive web interfaces using core technologies:

- HTML (Structure)
- CSS (Styling & Layout)
- JavaScript (Interactivity)

CORE FOCUS 03

Mobile Application Concepts

UI/UX design principles, event-driven mobile architecture, and cross-platform app creation.

CORE FOCUS 04

Game Development Systems

Physics engines, asset state management, game loops, and interactive user experiences.

CORE FOCUS 05

Robotics Integration

Connecting custom software pipelines to physical hardware, sensors, and motor controllers.

CORE FOCUS 06

AI Fundamentals & Prompts

Machine learning concepts, LLM interaction patterns, and systematic prompt engineering.

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Grades 7–8 Curriculum Stage — SOLVE & DEVELOP

Cognitive Objective: Understanding system architecture, complex data flow, and modern AI tool integration.

FOCUS AREA 01

Advanced Python & APIs

Advanced Python and API integration

FOCUS AREA 02

Data Systems

Database management and data flow

FOCUS AREA 03

App Engineering

Web and mobile application engineering

FOCUS AREA 04

AI & Vibe Coding

AI applications and vibe coding

FOCUS AREA 05

Problem Solving

Real-world problem solving methodologies

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Grade 9 Curriculum Stage — INNOVATE

CORE FOCUS AREAS

- **Software Architecture**
  Advanced algorithms and robust system design frameworks.
- **Autonomous AI Agents**
  Complex systems, decision trees, and intelligent behavior.
- **Web & Mobile Platforms**
  Advanced full-stack deployment and cross-platform solutions.
- **Independent Projects**
  End-to-end product development and technology entrepreneurship.

COGNITIVE OBJECTIVE

CAPSTONE MASTERY

End-to-End Creation

Mastering end-to-end product design, market application, and independent creation.

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Curriculum Domain Progression Matrix

Technology Domains Across Grades 1–9 • Mapping Progression from Introduction to Complete Mastery

TECHNOLOGY DOMAIN	GRADES 1–3	GRADES 4–6	GRADES 7–9
Computational Thinking	Logic	→	Architecture
Robotics & Tinkering	Mechanics	→	Autonomous Systems
Python	Visual Concepts	→	Advanced OOP
Web & Mobile Dev	UI Basics	→	Full-Stack Apps
Game Development	Sequencing	→	Physics Engines
AI & Vibe Coding	Basics & Prompting	→	Autonomous Agents
Entrepreneurship	Ideation	→	Market Launch

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The Python Progression Pathway

"We build the mental model before increasing technical complexity."

STEP 01

Foundation

- Logic
- Algorithms
- Visual Programming Concepts

STEP 02

Python Fundamentals

- Variables
- Conditions
- Loops
- Functions
- Lists

STEP 03

Intermediate

- Modules
- Files
- Errors
- Data
- APIs

STEP 04

Advanced

- OOP
- Databases
- Architecture
- Automation
- AI

STEP 05

Independent Dev

- Complete Systems
- AI Applications
- Production Products

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One Grade. Many Learning Journeys.

Prior Knowledge

Addressing prior knowledge and technical exposure

Learning Pace

Adapting to individual learning speed and confidence levels

Core Principle

"The grade defines the destination. The student's capability determines the route."

Interests & Strengths

Aligning with personal interests and cognitive strengths

Targeted Support

Identifying and supporting specific learning difficulties

Every student reaches the required milestone, but the pathway adapts to their unique profile.

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No Student Is Left Behind by the Starting Point

"We do not simply push students into advanced modules because of their age. We build the missing foundations and progressively move them forward."

STEP 01 • BRIDGE FRAMEWORK

Foundation

Fill baseline gaps in knowledge to ensure a stable learning platform before advancement.

→

STEP 02 • BRIDGE FRAMEWORK

Capability Mapping

Assess logic, problem-solving skills, and underlying learning difficulties accurately.

→

STEP 03 • BRIDGE FRAMEWORK

Personalized Path

Adapt learning speed and deliver targeted activities tailored to individual needs.

→

STEP 04 • BRIDGE FRAMEWORK

Progression

Align student growth directly with grade-level developmental targets and mastery.

Mentor, Not Just Instructor

The Active Role of a Make7 Mentor

Observe → Identify difficulties → Understand learning behavior → Personalize activities → Challenge → Track → Remould path

Key Monitoring Areas

- **Conceptual gaps** and recurring technical challenges
- **Individual strengths** and building confidence
- **Collaboration skills** and problem-solving habits

Mentors guide the cognitive journey, ensuring technical skills translate to deep understanding.



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From Student Ideas to Real Projects & Independent Creators

SUMMARY TRANSFORMATION ROADMAP

GRADE 1–2
Explore
→
GRADE 3–4
Think & Build
→
GRADE 5–6
Program & Create
→
GRADE 7–8
Solve & Develop
→
GRADE 9
Innovate

THE ENTREPRENEURIAL CYCLE

Problem → Research → Idea → Solution → Prototype → Development

Testing → Feedback → Improvement → Pitch → **Real Project**

“Thinking. Creating. Adapting.”